



ACTAI · WEEK 01 · LECTURE 03

Under the Hood

How AI actually works — from learning, to transformers, to predicting the next word.



A concept lecture — with a hands-on playground at the end

WHY PEEK INSIDE

You don't need a license to drive — but it helps to know what's under the hood



Demystify it

It isn't magic. A simple mental model makes AI far less intimidating.



Use it better

Knowing how it reads and predicts text explains why your prompts behave as they do.



Spot its limits

You'll see exactly why it can sound confident and still be completely wrong.

How we got here, fast



Each leap built on the last — but 2017 is the one that changed everything.

THE CORE IDEA

Machines learn from examples, not rules

Old way: a human writes every rule. New way: show the machine thousands of examples and let it find the patterns itself.

1

Lots of examples

e.g. millions of sentences,
images, or labels



2

Find the patterns

the model adjusts itself to fit
what it sees



3

Make predictions

now it can handle things it never
saw before

Supervised vs unsupervised



Supervised

Learning from labelled examples.

- You show it: 'this photo = cat', 'this = dog'
- It learns to label new photos itself
- Used for: spam filters, translation, predictions



Unsupervised

Finding patterns with no labels.

- You give it data with no answers attached
- It groups things that are similar on its own
- Used for: grouping customers, spotting odd data

LLMs learn mostly by a clever self-supervised trick: hide the next word and guess it — over and over.

The Transformer changed everything



Attention

It can weigh which words in a sentence matter most to each other — grasping context, not just word order.



Parallel & fast

It reads whole passages at once instead of word-by-word, so it trains far faster on huge data.



Scales up

Make it bigger + feed it more text = it keeps getting better. That's what produced today's LLMs.

*A 2017 paper titled “Attention Is All You Need” — the architecture behind GPT, Claude, Gemini, and the rest.
<https://arxiv.org/pdf/1706.03762>*

WHERE WE ARE NOW

The era of generative AI

Earlier AI

Recognised and sorted things.

“Is this email spam? Is this a cat?”

Generative AI

Creates brand-new content.

Text, images, audio, video, code — made on demand.

You are here. The tools you've been using all create — and they do it by predicting one piece of text at a time. Let's see how.



THE BIG SECRET

An LLM has one job: **predict the next word.**

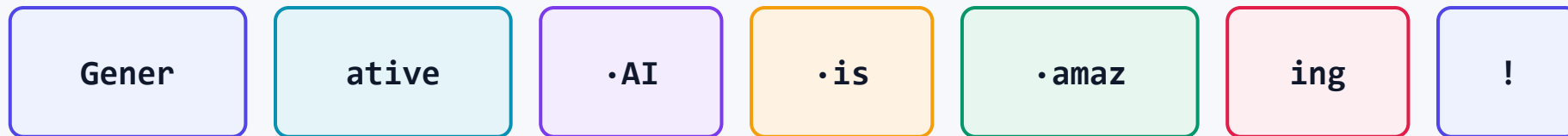
It isn't looking anything up, and it isn't thinking like you. It's guessing the next chunk of text — astonishingly well, billions of times.

STEP 1

Your words become tokens

Before anything else, your text is chopped into “tokens” — whole words or pieces of words.

“ **Generative AI is amazing!** ”



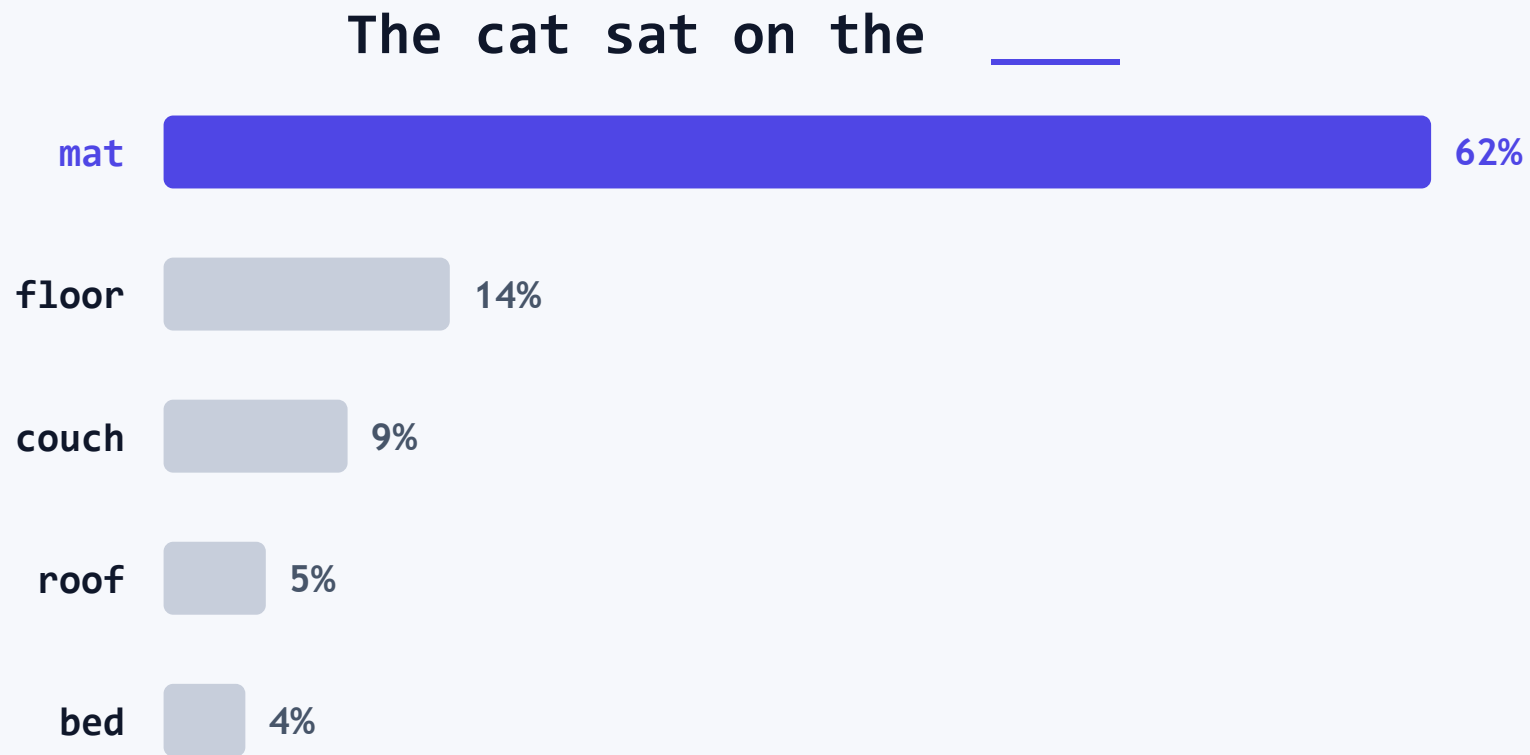
7 tokens. Notice words split into pieces — and the • dots are spaces, which count too.

Why it matters: the model never sees letters or 'words' the way you do — only tokens. It also explains why limits are counted in tokens, not words.

STEP 2

It predicts the next token

Given everything so far, it ranks every possible next token by how likely it is.



It usually picks the top one — but not always. That tiny bit of randomness is why the same prompt can give different answers.

STEP 3

...then it does it again, and again

It adds the chosen token, then predicts the next — one token at a time — until the answer is done.

The **cat**

The cat **sat**

The cat sat **on**

The cat sat on **the**

The cat sat on the **mat.**



**Predict → add →
repeat**

THE CATCH

Why it can be confidently wrong

If its only job is to predict plausible-sounding text, it will happily produce text that sounds right but isn't true. That's a hallucination.

No live facts

It isn't checking a database — it's predicting words. It can invent names, dates, and quotes.

Fluent ≠ correct

Confident, well-written text can still be wrong. Don't mistake polish for accuracy.

So: always verify

Ground it in text you provide, ask it to flag uncertainty, and check anything that matters.

This is the single most important thing to remember every time you use AI.



PLAYGROUND

Let's see it for ourselves.

Three quick experiments. Have a tool open — we'll do each one together, then you try it.

Be the model



Try it now



6 min

- Read a starter on the right out loud.
- Everyone calls out (or types) the next word.
- We tally the most common guesses.
- That's you doing next-token prediction!

FINISH THE SENTENCE

“I went to the ____”

“The best part of the weekend is ____”

“In the morning I always drink ____”

“She opened the door and saw a ____”

See your words as tokens



Try it now



8 min

- Open the tokenizer site on the right.
- Type or paste the sample sentence.
- Now paste the same idea in your own language.
- Compare the token counts — notice the difference.

OPEN IN YOUR BROWSER

tiktokenizer.app

ad-free backup: `gpt-tokenizer.dev`

PASTE THIS:

Generative AI is amazing!

Tip: English usually needs fewer tokens than other languages — try both.

Same prompt, different answers



Try it now



8 min

- Open any AI tool you like.
- Paste the prompt on the right.
- Generate, then hit regenerate 2–3 times.
- Compare — how different are they?

TRY THIS PROMPT

Write a 2-line poem about rain.

Then regenerate it 2–3 times and compare the results.

Why it changes: each run the model picks slightly different 'likely' tokens — the randomness from Step 2.

What you now understand



Modern AI took off around 2012, and the 2017 Transformer is what made today's tools possible.



Machines learn from examples — supervised (with labels) and unsupervised (finding patterns).



An LLM's core job is simply to predict the next token, over and over.



Your text becomes tokens first; the model only ever sees tokens.



Because it predicts plausible text rather than looking up facts, it can be confidently wrong — always verify.



THAT'S WEEK 1

You know what it is, and how it works.

Next week — Generating images, audio, and video. Time to create.